

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 0606

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

***CO1:** Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks: 25

Annexure A

CONCEPT MAPPING

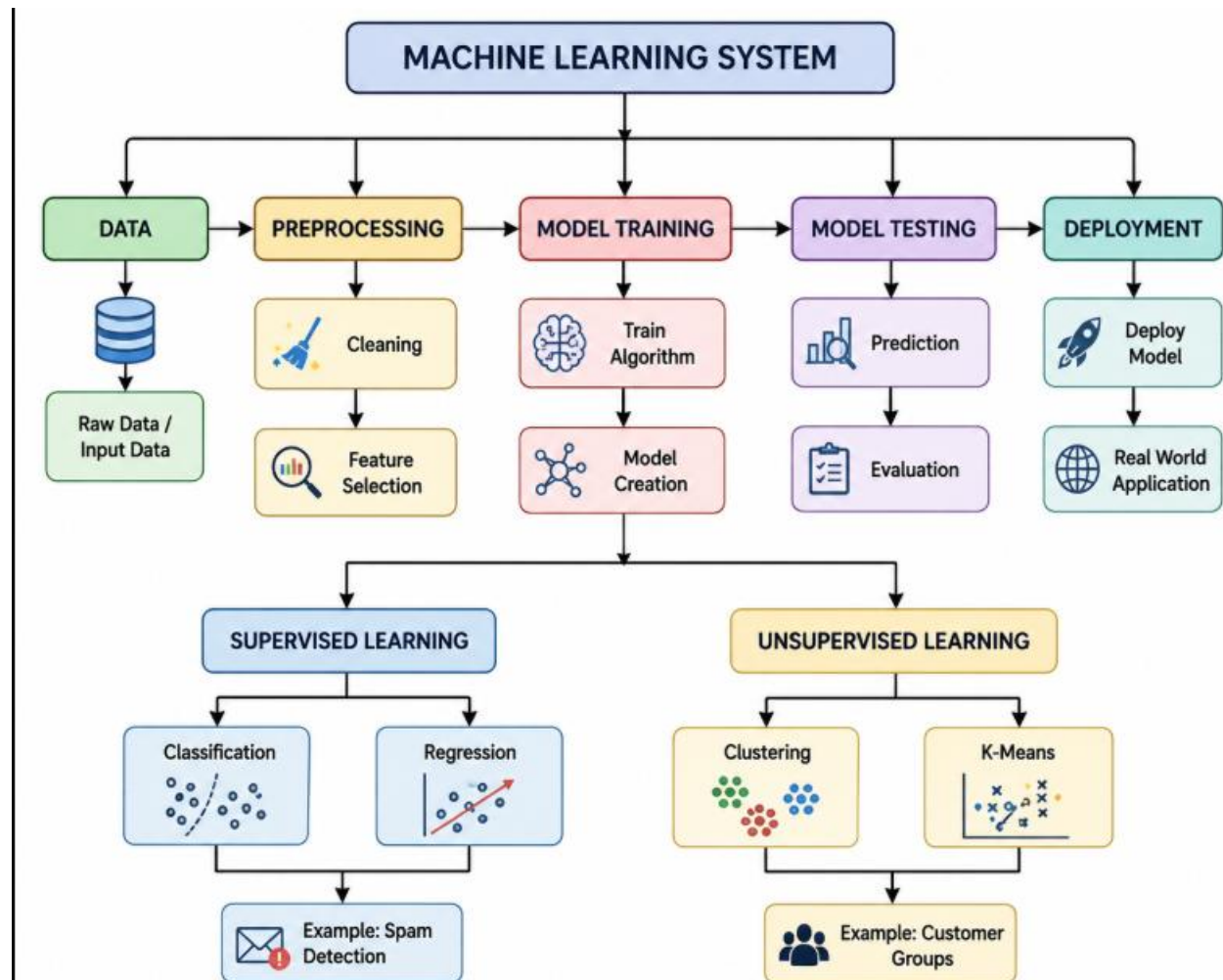
Code of Conduct:

I Deepan Kumar P (192312498) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

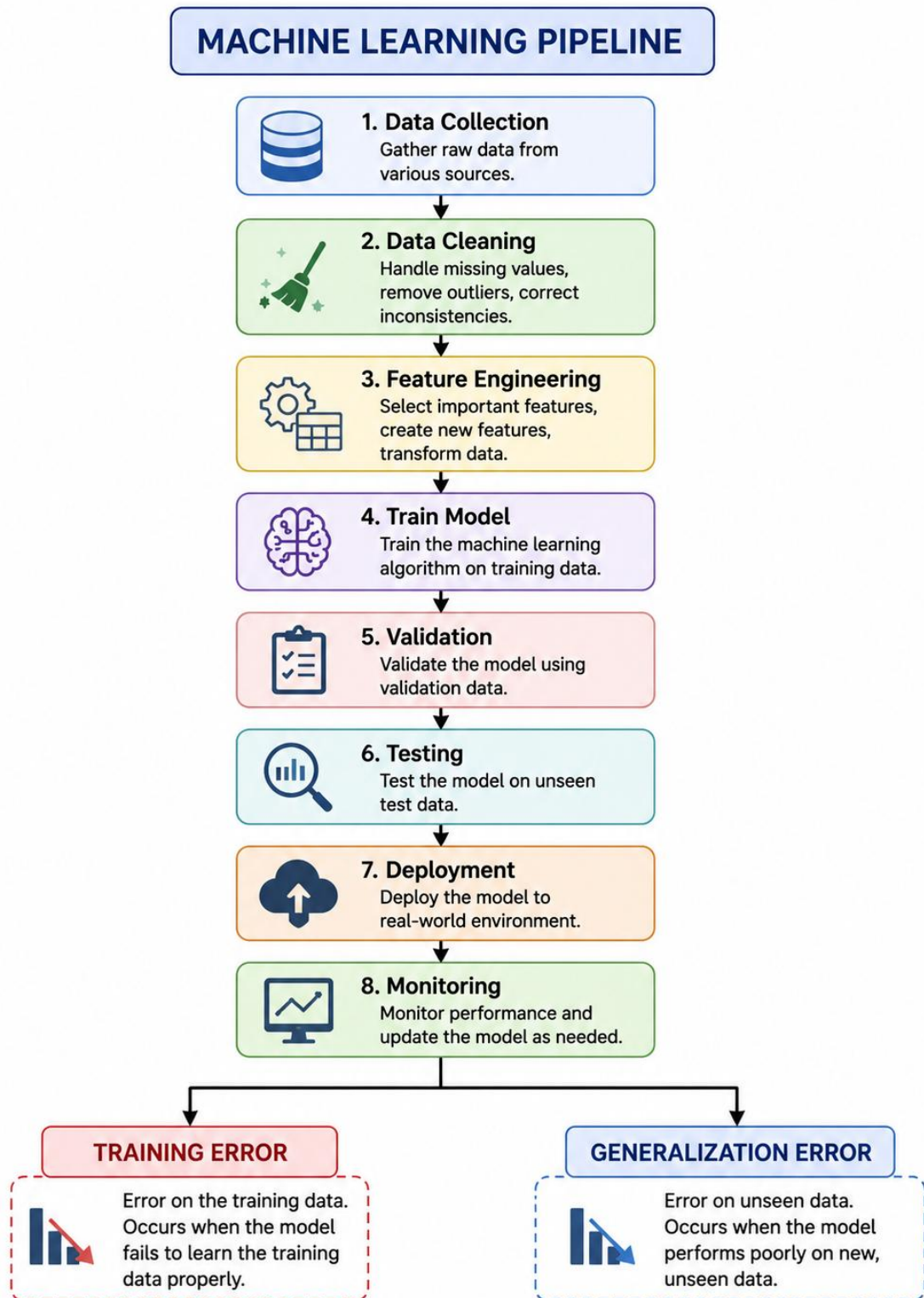
I understand that any violation of academic integrity rules will result in disciplinary action.

Deepan Kumar P

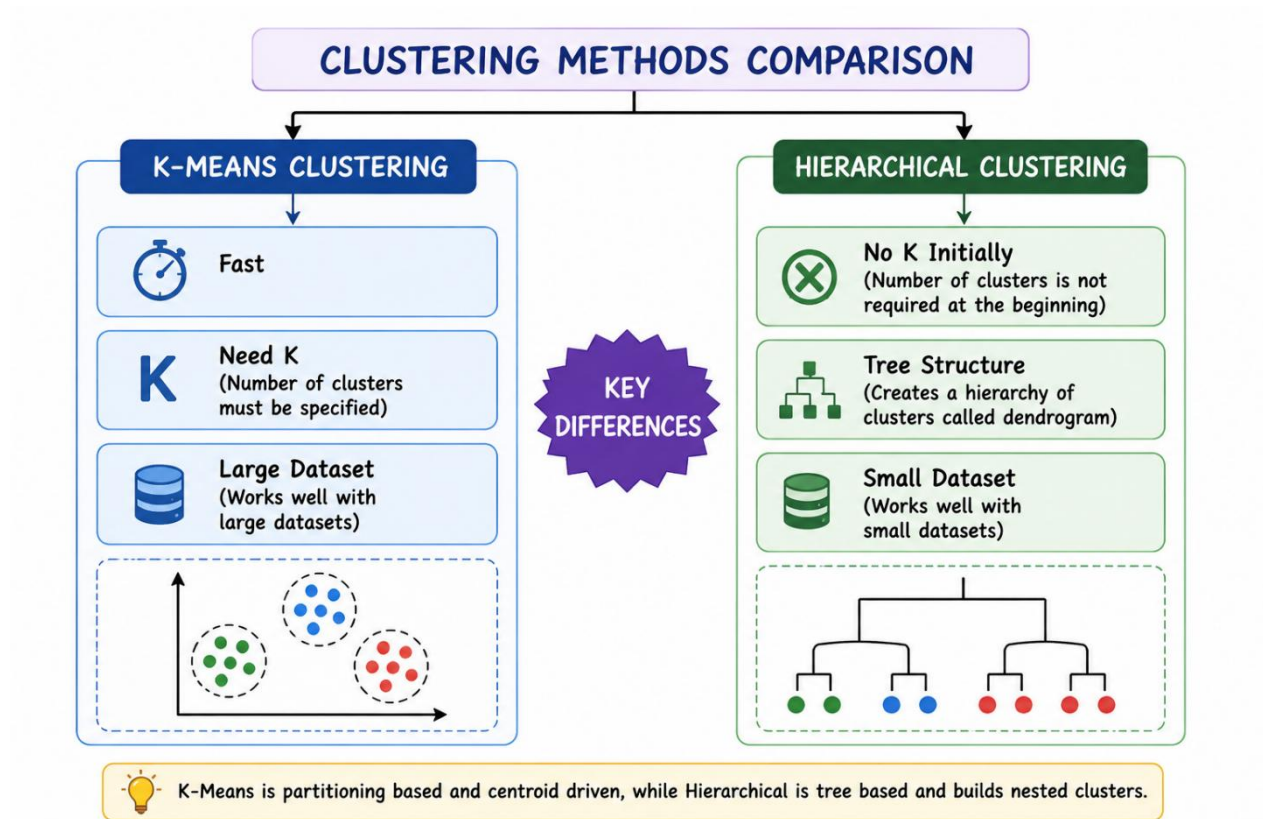
1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)



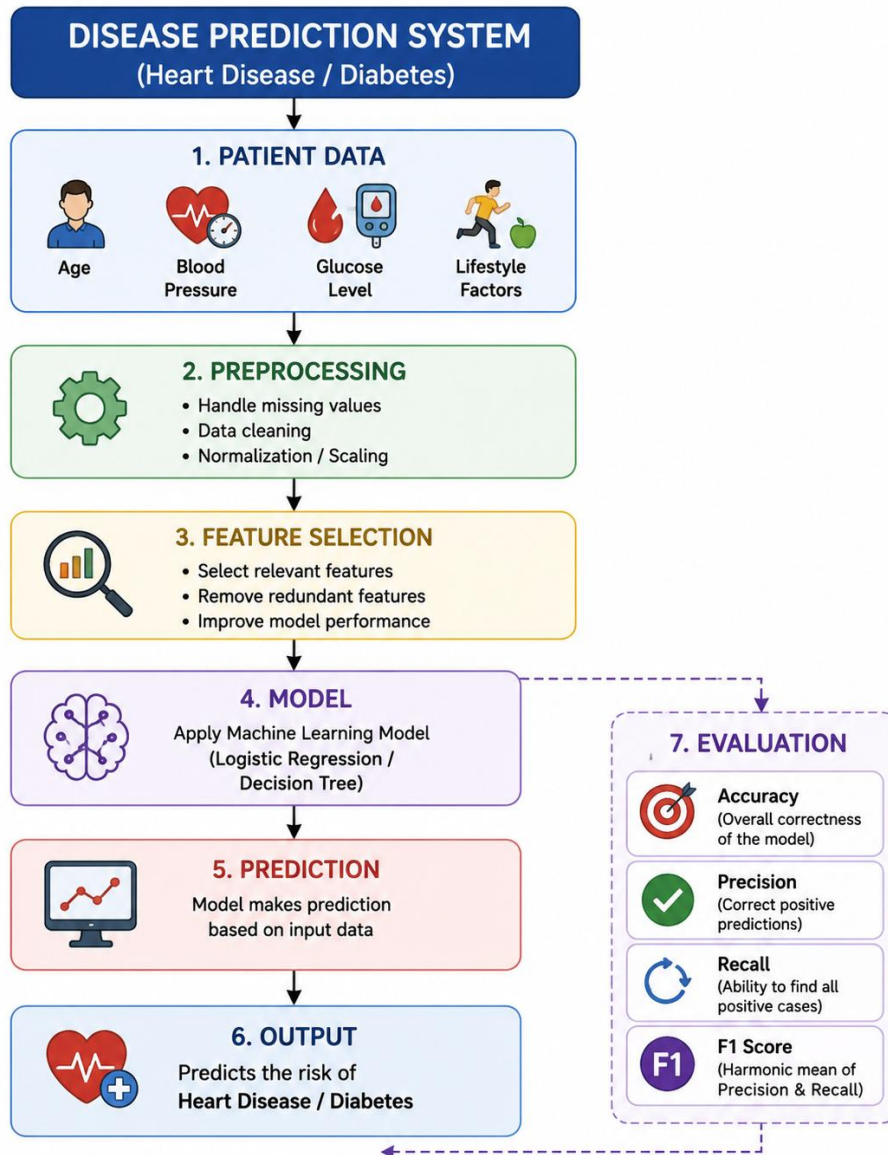
2. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)



3. Explain the concept of clustering in unsupervised learning. (5 Marks) (a) Describe the working of the K-Means algorithm with a step-by-step example. (b) Explain how the value of K is chosen using the Elbow Method. (c) Compare K-Means with Hierarchical Clustering.



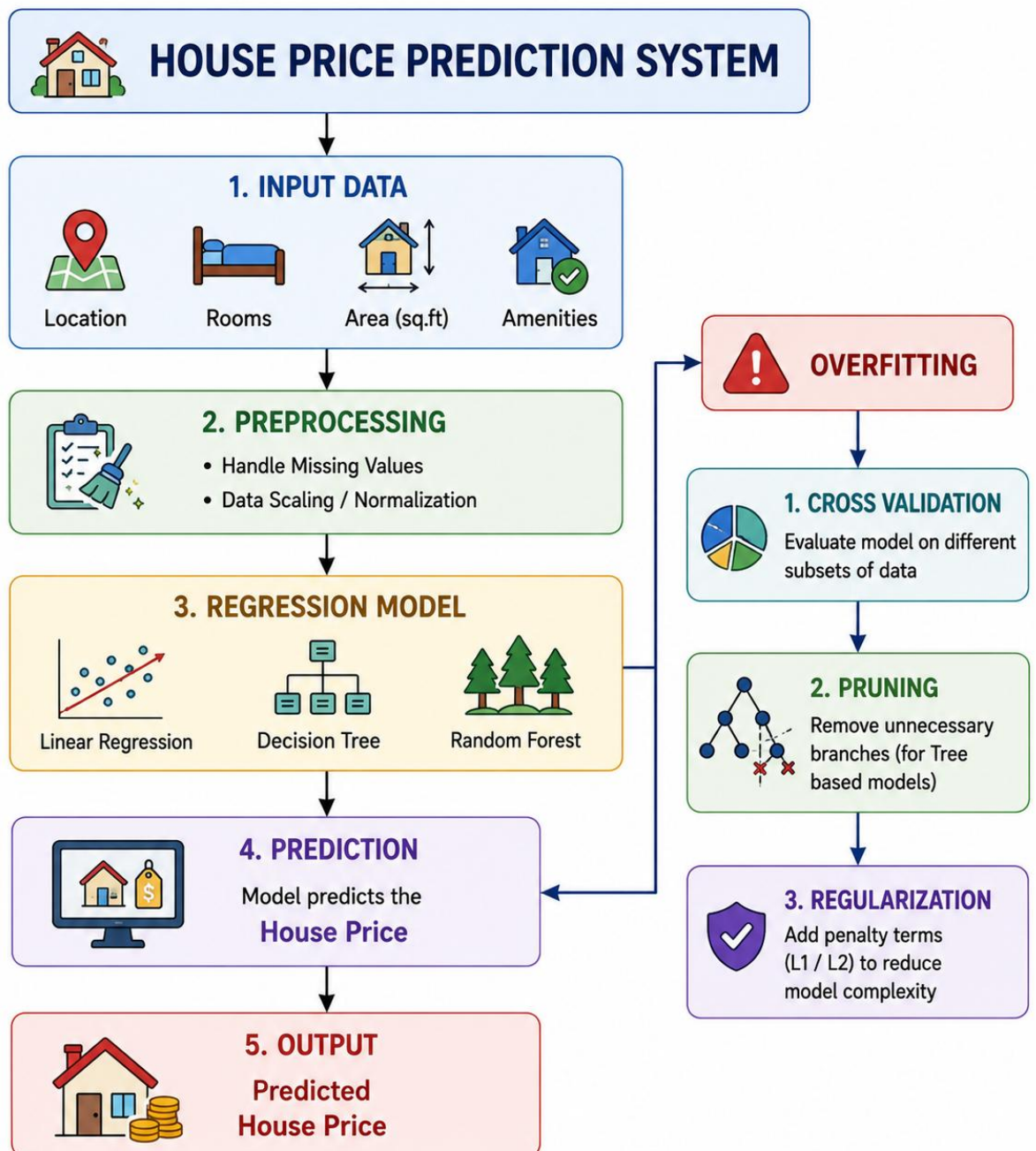
4.Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. (5 Marks)



5. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)

- Justify the choice of regression model (Linear Regression, Decision Tree, or RandomForest).
- Explain feature scaling and handling missing data.

(c) Describe how overfitting can be detected and avoided.



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand